

HF SIGNAL 01 — Architecture Note

Version 4, 27 August 2026 (supersedes 29 July 2026) · Giovanni Nardacci, Human Flag Association

Purpose

This note sets out, in one place, the architecture of HF SIGNAL 01 as currently published by the Association. It does not amend any deposited document, alter any identifier, or assert any obligation. It because the July 2026 working paper specifies one modality of the protocol in detail while describing the others in terms that no longer reflect the published architecture. A reader approaching the corpus from either direction should be able to identify the current architecture. HF SIGNAL 01 does not replace the white flag; it translates it. The gesture has not changed; the recipient has. A machine does not see a surrender: it detects, or fails to detect, a signal, and it shares no code with the white flag.

One protective function, two modalities

HF SIGNAL 01 is a single protective function expressed through two modalities with different levels of accessibility. The immediate operational response is the same in each case — do not engage; suspend and verify — although its legal significance depends on the circumstances. This note addresses only the physical protocol. It comprises two modalities:

The passive visual modality — a white disc on a red field. Its elementary geometry and strong chromatic contrast are intended to support low-resolution detection and rotational invariance. It requires no power, equipment, licence or emission. It is the modality available to a person who has nothing.

The active electronic modality — the emission specified in the working paper: a pulsating white light at 7 Hz combined with a continuous FM tone. It requires equipment and, with it, sufficient training to judge when emission is prudent.

The two modalities are alternative expressions of the same meaning; they are not cumulative requirements. Requiring both would exclude precisely those least equipped to signal. The visual symbol is therefore a current element of the protocol, not a superseded design iteration.

Legal effect, unchanged

The rule set out in the working paper applies to every modality without exception. A confirmed signal is sufficient to trigger immediate attack inhibition and status verification. It is never a necessary condition of protection, nor is it by itself conclusive proof of status. Its legal significance depends on the circumstances: for a civilian, it communicates or reinforces a protected status that exists independently of the signal; for a combatant, it may constitute a manifestation of an intention to surrender, subject to the conditions governing hors de combat status. In either case, the absence of the signal permits no adverse inference, and the full range of human-legible expressions of surrender retains its entire legal force. Any implementation that treated non-emission as an indicator of targetability would be inconsistent with applicable international humanitarian law, including the rules protecting civilians and, where surrender or hors de combat status is at issue, Article 41. The protocol is not a recognised emblem, sign or signal within the meaning of Article 38 of Additional Protocol I, and no such status is claimed for it.

Open questions

Carrier frequency. The designated carrier frequency and channel width require re-examination against the allocated amateur band, the applicable band plan and national licensing requirements.

Emission risk. A periodic radio emission creates a direction-finding opportunity. The conditions in which emission is advisable, and whether the radio modality should be restricted to equipped and trained emitters, remain to be specified.

Photosensitivity. The optical channel pulses at a frequency above the threshold conventionally treated as safe for photosensitive individuals; general guidance on flashing light places that threshold at three flashes per second. A duty cycle, a maximum emission duration, an intensity ceiling or a lower pulse frequency must therefore be specified. Each choice affects detectability. No assessment has yet been carried out.

Separability. The visual symbol must be shown to be reliably distinguishable by machines, at low resolution, from the protective emblems of the Geneva Conventions.

Symbol parameters. Chromaticity, the disc-to-field ratio, minimum size, viewing angle and infrared behaviour remain to be specified.

Category payload. It remains an open question whether the symbol should include a printable fiducial marker carrying a category, but no personal identifier.

A printed marker can be forged. So can the white flag, and the law has responded by prohibiting perfidy rather than abandoning the sign. No claim of unforgeability is made here.

Status

Working document of the Human Flag Association. It supplements, but does not amend, the HF SIGNAL 01 working paper (July 2026) or any deposited item in the corpus. Its contents are intended for the next revision of that paper. This note is deposited on Zenodo under concept DOI 10.5281/zenodo.21809826; the HF SIGNAL 01 working paper (July 2026), which it supplements, is deposited under concept DOI 10.5281/zenodo.21183137. Concept DOIs are used throughout and resolve to the current version.